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Exp-16: Compare different types Classification Algorithms and evaluate their performance.

Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.tree import DecisionTreeClassifier

from sklearn.neighbors import KNeighborsClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score

Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

Split the dataset

X_train, X_test, y_train, y_test = train_test_split(

 X, y, test_size=0.2, random_state=42

)

Create models

models = {

 "Logistic Regression": LogisticRegression(max_iter=200),

 "Decision Tree": DecisionTreeClassifier(random_state=42),

 "KNN": KNeighborsClassifier(n_neighbors=3),

 "Naive Bayes": GaussianNB()

}

print("Algorithm\t\tAccuracy")

Train and evaluate each model

```
for name, model in models.items():  
    model.fit(X_train, y_train)  
    y_pred = model.predict(X_test)  
    accuracy = accuracy_score(y_test, y_pred)  
    print(f"{name:20} {accuracy*100:.2f}%")
```

Output:

```
... Algorithm      Accuracy  
  Logistic Regression 100.00%  
  Decision Tree      100.00%  
  KNN                 100.00%  
  Naive Bayes        100.00%
```